

Solutions and R Programming Examples: Jackknife and Bootstrap methods

1. Fuel efficiency

```
# data
```

```
fuel<-c(26,23,21,23,19,29,15,26,19,26)
```

```
# Sample mean
```

```
mean(fuel)
```

```
.....  
[1] 22.7  
.....
```

```
library(bootstrap)
```

```
# Jackknife the mean
```

```
jackmean <- jackknife(fuel,mean)
```

```
jackmean
```

```
.....  
$jack.se
```

```
[1] 1.342055
```

```
$jack.bias
```

```
[1] 0
```

```
$jack.values
```

```
[1] 22.33333 22.66667 22.88889 22.66667 23.11111 22.00000 23.55556 22.33333
```

```
[9] 23.11111 22.33333
```

```
$call
```

```
jackknife(x = fuel, theta = mean)
```

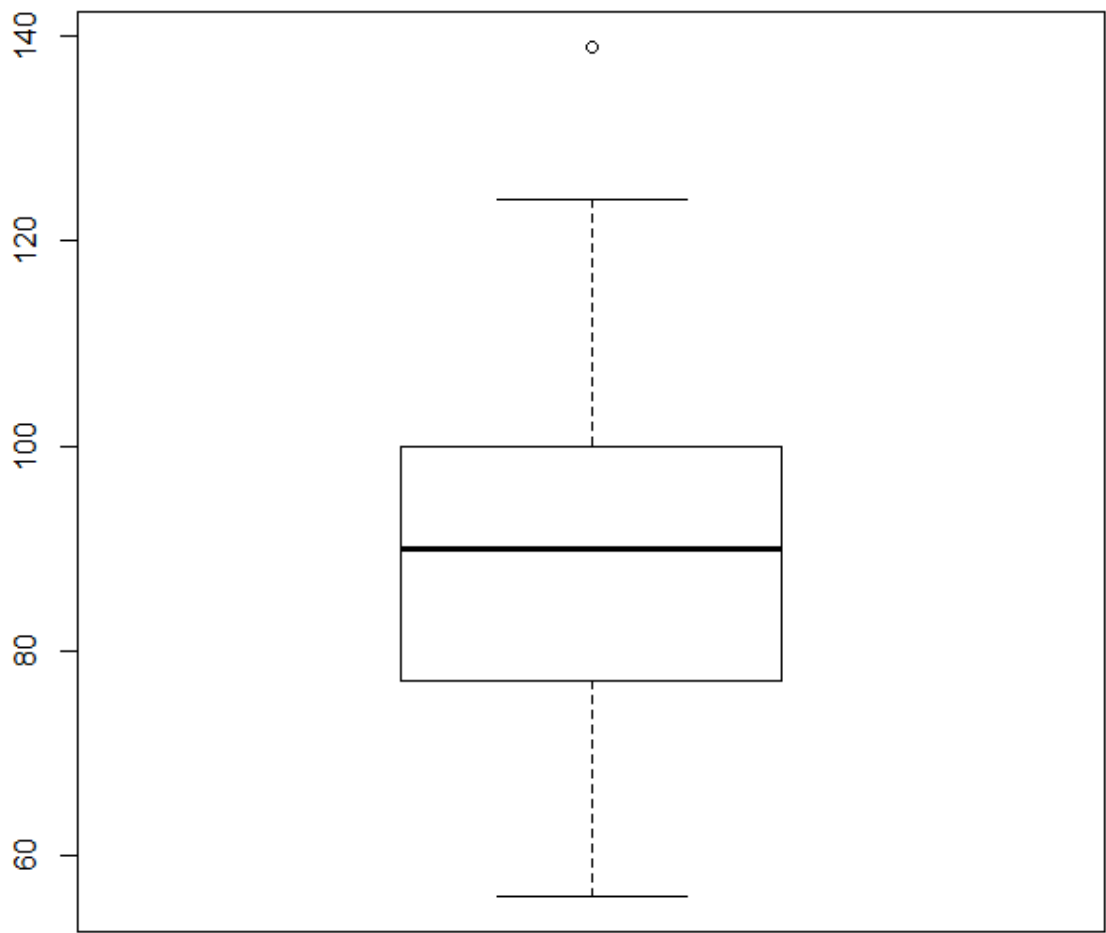
```
# Bias-corrected jackknife estimate  
meanjack = mean(fuel) - jackmean$jack.bias  
meanjack
```

```
.....  
[1] 22.7  
.....
```

2.Family income

```
# data  
faminc <- c(90, 77, 100, 83, 64, 78, 92, 73, 122, 96, 60, 85, 86, 108, 70, 139, 56, 94, 84, 111,  
93, 120, 70, 92, 100, 124, 59, 112, 79)  
boxplot(faminc)
```

```
.....
```



.....

```
t.test(faminc)
```

One Sample t-test

data: faminc

t = 23.149, df = 28, p-value < 2.2e-16

alternative hypothesis: true mean is not equal to 0

95 percent confidence interval:

82.25622 98.22654

sample estimates:

mean of x

90.24138

```
library(boot)
```

```
set.seed(11)
```

```
mean.fun <- function(faminc, i)
```

```
  {m <- mean(faminc[i]) }
```

```
b <- boot(faminc, mean.fun, R = 2000)
```

```
b
```

ORDINARY NONPARAMETRIC BOOTSTRAP

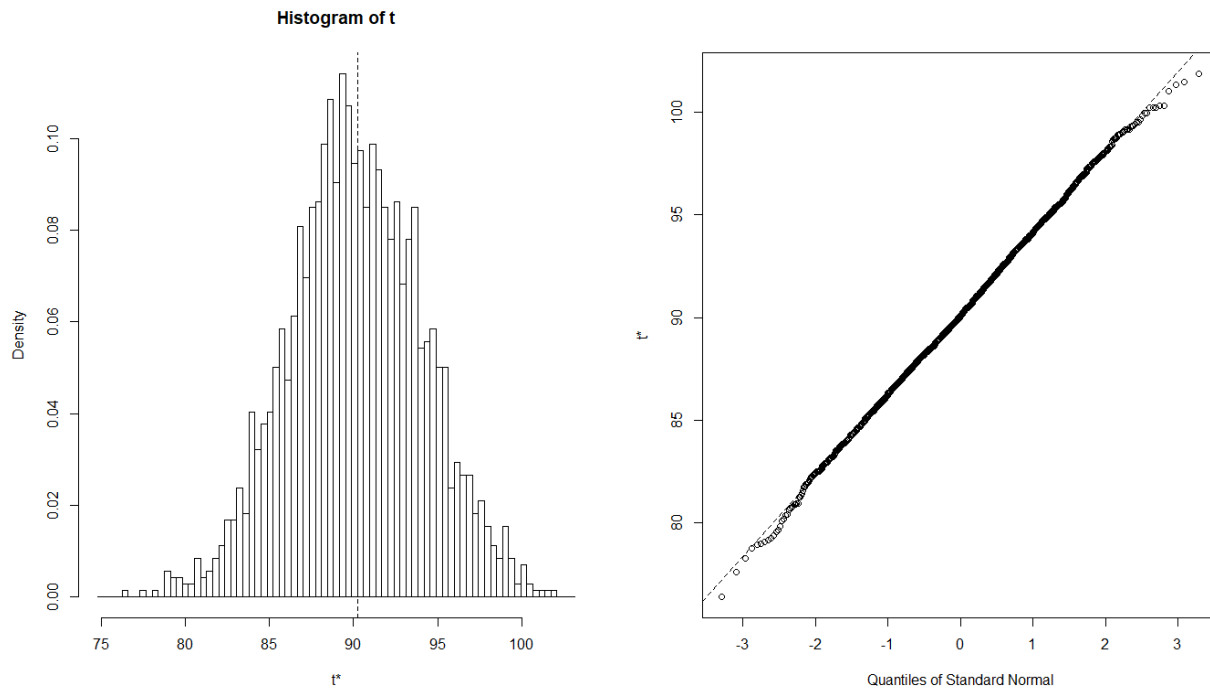
Call:

```
boot(data = faminc, statistic = mean.fun, R = 2000)
```

Bootstrap Statistics :

original	bias	std. error
t1* 90.24138	-0.08084483	3.935964

plot(b)



```
ci<-boot.ci(b, conf = 0.95, type = c("norm" ,"perc", "bca"))
```

ci

BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS

Based on 2000 bootstrap replicates

CALL :

```
boot.ci(boot.out = b, conf = 0.95, type = c("norm", "perc", "bca"))
```

Intervals :

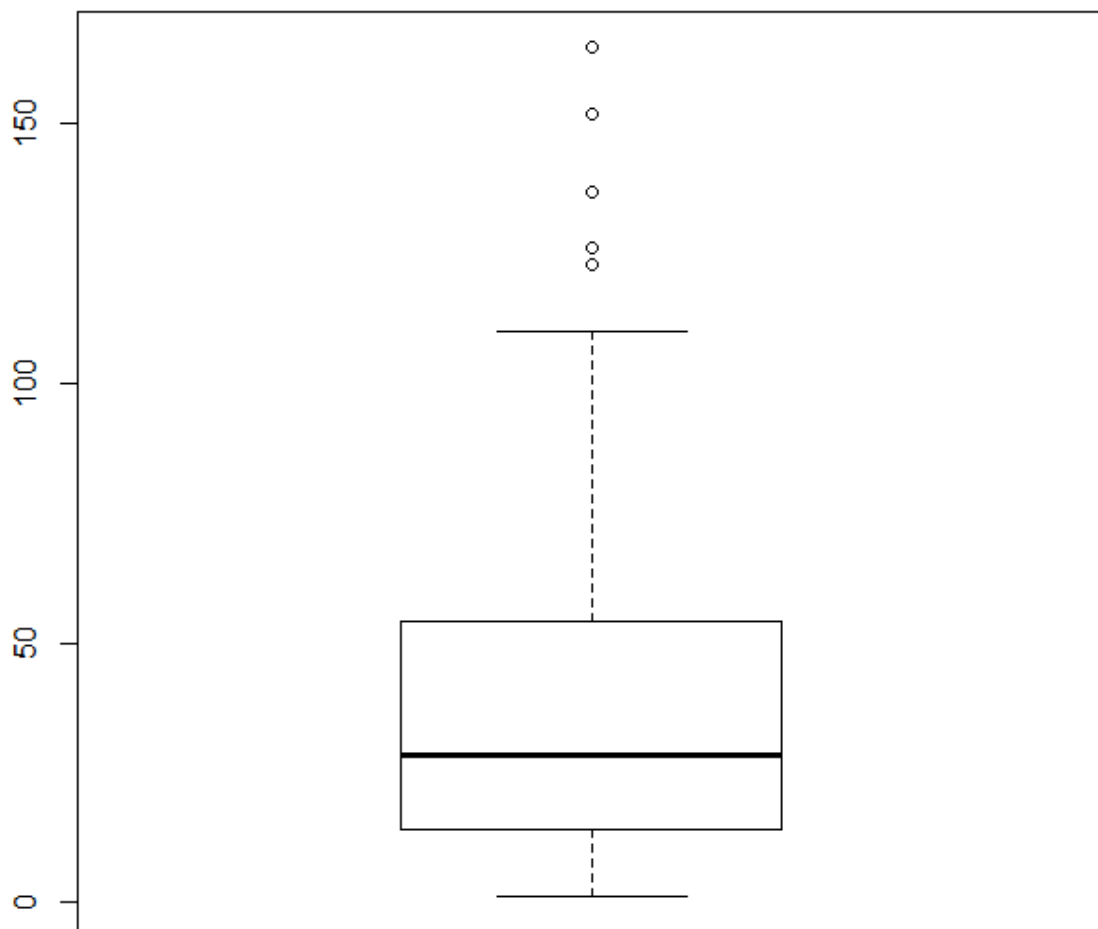
Level	Normal	Percentile	BCa
95%	(82.61, 98.04)	(82.52, 97.90)	(82.90, 98.33)

Calculations and Intervals on Original Scale

3. Random samples 50 objects

```
x<- c(54,5, 35, 137, 31, 27, 152, 2, 123, 81, 74, 27, 11, 19, 126, 110, 110, 29, 61, 35, 94, 31,  
26, 5, 12, 4, 165, 32, 29, 28, 29, 26, 25, 1, 14, 13, 13, 10, 5, 27, 4, 52, 30, 22, 36, 26, 20,  
23, 33, 68)
```

```
boxplot(x)
```



```
t.test(x)
```

One Sample t-test

data: x

t = 7.2589, df = 49, p-value = 2.637e-09

alternative hypothesis: true mean is not equal to 0

95 percent confidence interval:

31.12475 54.95525

sample estimates:

mean of x

43.04

```
library(boot)
```

```
set.seed(11)
```

```
mean.fun <- function(x, i)
```

```
  {m <- mean(x[i]) }
```

```
b2 <- boot(x, mean.fun, R = 2000)
```

```
b2
```

ORDINARY NONPARAMETRIC BOOTSTRAP

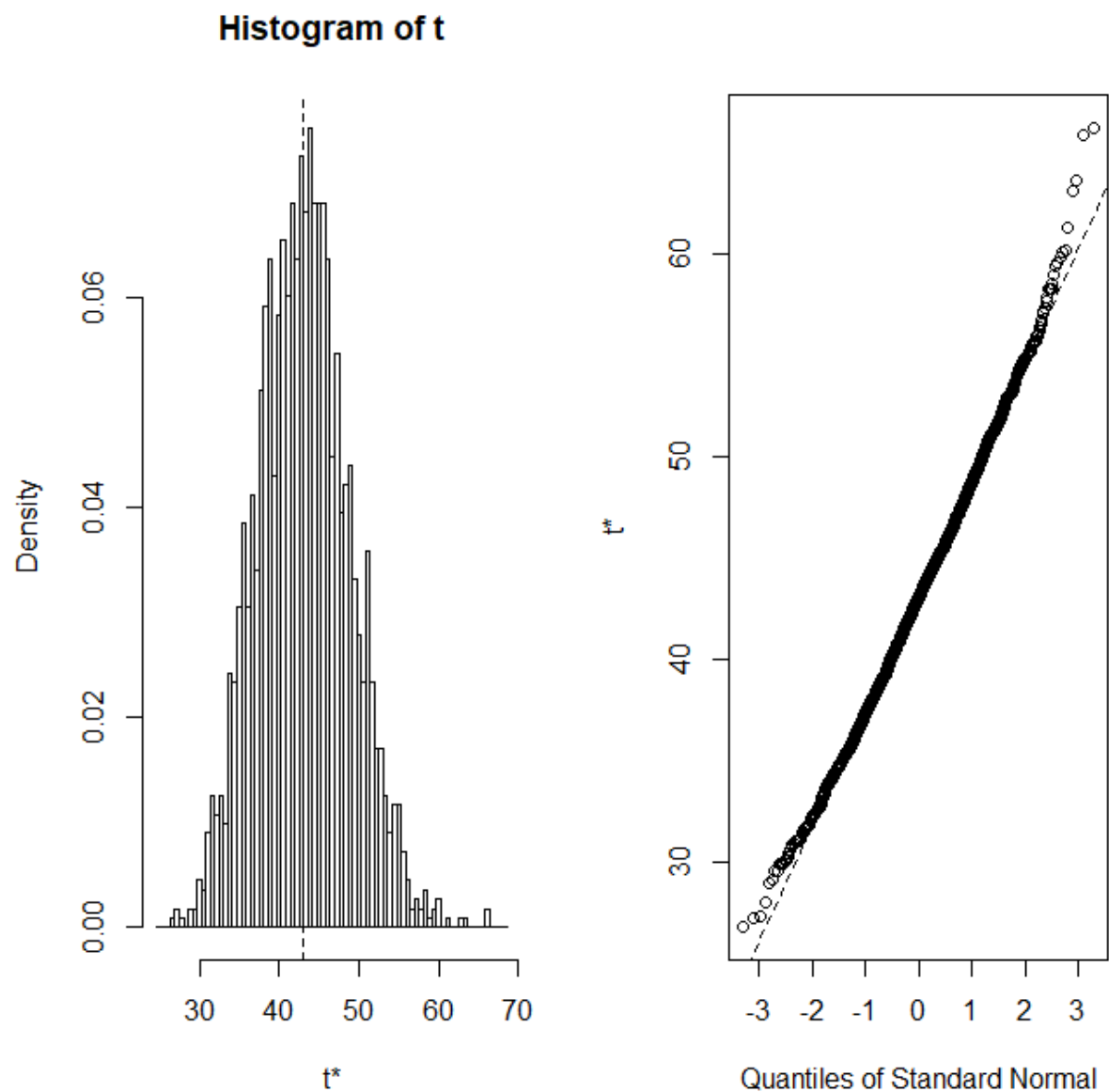
Call:

```
boot(data = x, statistic = mean.fun, R = 2000)
```

Bootstrap Statistics :

	original	bias	std. error
t1*	43.04	0.04051	5.698286

```
plot(b2)
```



```
ci<-boot.ci(b2, conf = 0.95, type = c("norm", "perc", "bca"))
```

```
ci
```

BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS

Based on 2000 bootstrap replicates

CALL :

```
boot.ci(boot.out = b, conf = 0.95, type = c("norm", "perc", "bca"))
```

Intervals :

Level	Normal	Percentile	BCa
95%	(31.83, 54.17)	(32.32, 54.68)	(32.81, 55.12)

Calculations and Intervals on Original Scale
